

Qualification Pack



Waste Optimisation Professional

QP Code: SGJ/Q5002

Version: 1.0

NSQF Level: 6

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SGJ/Q5002: Waste Optimisation Professional

Brief Job Description

India is facing major environmental challenges associated with waste generation and inadequate waste collection, transport, treatment and disposal. Current systems in India cannot cope with the volumes of waste generated by an increasing urban population, and impact on the environment and public health. The course will facilitate management of various categories of waste (e.g., solid waste, plastic, biomedical waste, e-waste, C&D waste etc.) including source segregation, collection, disposal, processing and recycling.

Personal Attributes

This job requires the individual to be diligent, organised, punctual, attentive and accountable to complete the work safely and timely.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4072: Introduction to Solid Waste](#)
2. [SGJ/N4073: Identify characteristic, collection, transfer and transport and Separation and processing of solid waste](#)
3. [SGJ/N4074: Perform Composting and recycling, waste to energy and sanitary landfilling](#)
4. [SGJ/N4075: Introduce climate change in context to waste.](#)
5. [SGJ/N4076: Plan and develop strategy for waste management focusing circular economy and plastic waste management](#)
6. [SGJ/N4077: Perform E-Waste Management](#)
7. [SGJ/N4078: Perform Construction and Demolition activities within waste management frameworks, alongside addressing Bio-Medical Waste Management concerns and explores the fundamental principles of Geographic Information Systems applied to waste management practices.](#)
8. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
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Sub-Sector	Waste Management
Occupation	Entrepreneur
Country	India
NSQF Level	6
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO 2015 2143.0400 - Waste Management Engineer , Radioactive material NCO 2015 2263.9900 - Environmental and Occupational Health and hygiene Professionals, Other
Minimum Educational Qualification & Experience	Completed 4 year UG program (in relevant field) with 1 Year of experience of relevant experience OR Completed 3 year UG degree (in relevant field) with 2 Years of experience of relevant experience OR 12th Class (+2 year of diploma in relevant field) with 3 Years of experience of relevant experience OR Previous relevant Qualification of NSQF Level (5.5) with 1-2 Years of experience with 1.5 years of relevant experience OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience of relevant experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-06-ES-01755-2023-V1-SCGJ
NQR Version	1.0



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Remarks:

Total 570 hours (Theory: 120 hours Practical: 210 hours OJT : 150 hours Employability skills: 90 hours)

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SGJ/N4072: Introduction to Solid Waste

Description

This unit tells about Introduction to Solid Waste

Scope

The scope covers the following :

- Learn about solid waste Management and understand problems and challenges associated with the generation of solid waste.

Elements and Performance Criteria

Introduction to Solid Waste

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about understanding of challenges associated with generation of waste.
- PC2.** explain health and environmental impacts of poor waste management.
- PC3.** discuss elements of a Solid Waste Management System.
- PC4.** explain integrated Solid Waste Management.
- PC5.** show different types of waste collection through pictures or videos.
- PC6.** showcase challenges associated with generation of waste.
- PC7.** showcase Health and environmental impacts of poor waste management.
- PC8.** show how to implement integrated solid waste management at site.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding the composition and types of waste generated (organic, plastic, paper, metal, glass, etc.) is essential for designing effective management strategies.
- KU2.** minimizing waste generation at the source is the most sustainable approach. This involves promoting practices like reusable products, packaging reduction, and composting food scraps.
- KU3.** efficient collection systems ensure timely waste removal, preventing accumulation and environmental hazards. Different methods like curbside collection, drop-off centers, and community composting can be employed.
- KU4.** different technologies are used to process waste, depending on its type and composition. These include recycling, composting, incineration, and landfilling.
- KU5.** landfills, though necessary for non-recyclable waste, pose environmental risks if not managed properly. Practices like leachate collection, methane capture, and proper closure are crucial.
- KU6.** improper SWM can lead to air and water pollution, soil contamination, and the spread of diseases. Understanding these impacts is vital for promoting responsible waste management.
- KU7.** SWM has social and economic implications, including job creation, resource recovery, and public health costs. Effective management requires considering these aspects.

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- KU8.** laws and regulations govern waste management practices, ensuring environmental protection and public health. Understanding these frameworks is essential for compliance.
- KU9.** public awareness and active participation are critical for successful SWM. Educational campaigns and community engagement initiatives can promote responsible waste disposal behaviors.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** understanding federal, state, and local laws and regulations pertaining to solid waste collection, transportation, disposal, and recycling.
- GS2.** familiar with the different types of solid waste, such as municipal solid waste, commercial and industrial waste, construction and demolition waste, and hazardous waste.
- GS3.** understanding how landfills, incinerators, composting facilities, and other solid waste management facilities operate.
- GS4.** able to collect, analyze, and interpret data on solid waste generation, collection, and disposal.
- GS5.** able to identify and solve problems related to solid waste management, such as illegal dumping, littering, and
- GS6.** this includes being able to communicate effectively with a variety of stakeholders, such as government officials, businesses, and the public.
- GS7.** this includes being able to work effectively with others in a team environment.
- GS8.** this includes being able to give presentations and speeches about solid waste management to a variety of audiences.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solid Waste</i>	25	25	-	-
PC1. discuss about understanding of challenges associated with generation of waste.	5	-	-	-
PC2. explain health and environmental impacts of poor waste management.	6	-	-	-
PC3. discuss elements of a Solid Waste Management System.	7	-	-	-
PC4. explain integrated Solid Waste Management.	7	-	-	-
PC5. show different types of waste collection through pictures or videos.	-	7	-	-
PC6. showcase challenges associated with generation of waste.	-	7	-	-
PC7. showcase Health and environmental impacts of poor waste management.	-	6	-	-
PC8. show how to implement integrated solid waste management at site.	-	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4072
NOS Name	Introduction to Solid Waste
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4073: Identify characteristic, collection, transfer and transport and Separation and processing of solid waste

Description

This unit tells about Identify characteristic, collection, transfer and transport and Separation and processing of solid waste. The course will facilitate management of various categories of waste (e.g., solid waste, plastic, biomedical waste, e-waste, C&D waste etc.) including source segregation, collection, disposal, processing and recycling.

Scope

The scope covers the following :

- understand the sources type and composition of solid waste and classification of solid waste such as physical, biological, economic and ecological.
- Learn about how to design a solid waste collection system including transfer and transport and Separation and processing of solid waste.
- Understand concept of a material recovery facility (MRF) and importance of health and hygiene

Elements and Performance Criteria

Characteristics of Solid Waste

To be competent, the user/individual on the job must be able to:

- PC1.** depict the sources, types and composition of solid waste.
- PC2.** identify quantities of solid waste.
- PC3.** demonstrate properties of solid waste.
- PC4.** perform data collection on waste.
- PC5.** understand how different materials, product design, and consumer choices contribute to waste generation.
- PC6.** identify strategies for source reduction like minimizing packaging, choosing durable products, and composting food scraps.

Collection, transfer and transport

To be competent, the user/individual on the job must be able to:

- PC7.** show how to develop a solid waste collection system.
- PC8.** show types of solid waste storage systems and collection vehicles.
- PC9.** show design of solid waste collection systems.
- PC10.** illustrate basic concepts, types and design of solid waste transfer stations.
- PC11.** perform and understand different collection methods (curbside, centralized, etc.) and select the most effective one for your needs.
- PC12.** show how to consider waste composition and geography when designing routes and optimizing resource allocation.
- PC13.** show how to utilize data analysis to track collection performance and identify areas for improvement.

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Separation and processing

To be competent, the user/individual on the job must be able to:

- PC14.** discuss basic concepts of separation and processing of solid waste.
- PC15.** discuss basic concepts of operation of a Material Recovery Facility.
- PC16.** explain design of a Material Recovery Facility.
- PC17.** perform the different stages of waste processing at an MRF, such as sorting, separation, and baling.
- PC18.** demonstrate the various technologies employed in MRFs, such as conveyor belts, trommels, and optical sorters.
- PC19.** identify the challenges and opportunities associated with MRF operation, such as contamination and market fluctuations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** landfills are used for disposing of non-recyclable and non-treatable waste. Minimizing reliance on landfills is essential due to environmental concerns.
- KU2.** technologies like gasification and plasma gasification convert waste into usable energy sources.
- KU3.** this approach aims to minimize waste generation by promoting product design for reuse, repair, and recycling, creating a closed-loop system.
- KU4.** solid waste comprises diverse materials like organic matter, paper, plastics, glass, metals, textiles, and electronic waste. Understanding the proportions of these components is crucial for designing effective management strategies.
- KU5.** different waste materials possess varying properties like density, moisture content, flammability, and reactivity. These properties influence storage, transportation, processing, and disposal methods.
- KU6.** organic waste degrades naturally, while others like plastics persist for extended periods. Biodegradability is essential for composting and other biological treatment processes.
- KU7.** separating waste at the source (homes, offices, etc.) into categories like recyclables, compostables, and general waste streamlines processing and enhances recycling efficiency.
- KU8.** depending on waste type, volume, and local infrastructure, various collection methods are employed, including curbside pickup, drop-off centers, and communal bins.
- KU9.** these facilities consolidate waste from smaller collection vehicles into larger ones for efficient long-distance transportation to processing or disposal sites.
- KU10.** techniques like screening, air separation, and magnetic separation segregate waste components based on size, density, and material properties.
- KU11.** composting biodegrades organic waste into nutrient-rich compost for soil amendment.
- KU12.** processes like anaerobic digestion convert organic waste into biogas, a renewable energy source.
- KU13.** incineration burns waste at high temperatures to reduce volume and generate energy. However, air pollution control is crucial.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** adapting to different waste streams, identifying inefficiencies, and developing solutions for complex challenges.
- GS2.** working effectively with diverse stakeholders, including residents, businesses, government agencies, and NGOs.
- GS3.** handling multiple tasks and deadlines efficiently, particularly in fast-paced environments.
- GS4.** understanding trends and patterns in waste generation to inform decision-making.
- GS5.** creating and delivering clear, engaging information about waste reduction, recycling, and responsible disposal.
- GS6.** staying updated on relevant laws and policies governing waste management.
- GS7.** knowing the benefits and drawbacks of various composting, recycling, and disposal methods.
- GS8.** safely using specialized machinery for collection, separation, and processing.
- GS9.** planning, implementing, and evaluating waste management initiatives.
- GS10.** ensuring efficient use of resources and securing funding for projects.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Characteristics of Solid Waste</i>	-	14	-	-
PC1. depict the sources, types and composition of solid waste.	-	2	-	-
PC2. identify quantities of solid waste.	-	2	-	-
PC3. demonstrate properties of solid waste.	-	2	-	-
PC4. perform data collection on waste.	-	2	-	-
PC5. understand how different materials, product design, and consumer choices contribute to waste generation.	-	3	-	-
PC6. identify strategies for source reduction like minimizing packaging, choosing durable products, and composting food scraps.	-	3	-	-
<i>Collection, transfer and transport</i>	-	14	-	-
PC7. show how to develop a solid waste collection system.	-	2	-	-
PC8. show types of solid waste storage systems and collection vehicles.	-	2	-	-
PC9. show design of solid waste collection systems.	-	2	-	-
PC10. illustrate basic concepts, types and design of solid waste transfer stations.	-	2	-	-
PC11. perform and understand different collection methods (curbside, centralized, etc.) and select the most effective one for your needs.	-	2	-	-
PC12. show how to consider waste composition and geography when designing routes and optimizing resource allocation.	-	2	-	-
PC13. show how to utilize data analysis to track collection performance and identify areas for improvement.	-	2	-	-
<i>Separation and processing</i>	16	7	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. discuss basic concepts of separation and processing of solid waste.	5	-	-	-
PC15. discuss basic concepts of operation of a Material Recovery Facility.	5	-	-	-
PC16. explain design of a Material Recovery Facility.	6	-	-	-
PC17. perform the different stages of waste processing at an MRF, such as sorting, separation, and baling.	-	2	-	-
PC18. demonstrate the various technologies employed in MRFs, such as conveyor belts, trommels, and optical sorters.	-	2	-	-
PC19. identify the challenges and opportunities associated with MRF operation, such as contamination and market fluctuations.	-	3	-	-
NOS Total	16	35	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4073
NOS Name	Identify characteristic, collection , transfer and transport and Separation and processing of solid waste
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N4074: Perform Composting and recycling, waste to energy and sanitary landfilling

Description

This unit tells about Perform Composting and recycling, waste to energy and sanitary landfilling

Scope

The scope covers the following :

- Understand the concept of solid waste composting and explain the technologies associated with solid waste composting.
- Learn basic concept and technologies related to energy recovery from solid waste and understanding refused drive fuel (RDF).
- Understand concept of sanitary landfills and learn different type of landfill and landfilling method and landfill leachate control and treatment.

Elements and Performance Criteria

Composting and recycling

To be competent, the user/individual on the job must be able to:

- PC1.** show types of solid waste recyclable materials.
- PC2.** prepare design and implementation of a recycling programme.
- PC3.** demonstrate basic concepts of solid waste composting.
- PC4.** show composting technologies and approaches.
- PC5.** show how to develop a solid waste composting programme.
- PC6.** perform Bio-gasification of solid waste.
- PC7.** show how to master the different types of recyclable materials like paper, metal, glass, plastic, and organics.
- PC8.** show how to learn how to implement an effective recycling program in your home, community, or organization.
- PC9.** ensure understanding the basics principles of composting, the breakdown of organic waste into nutrient-rich soil amendment.
- PC10.** show how to discover various composting methods like bin composting, vermicomposting, and windrow composting.
- PC11.** show how to equip yourself with the knowledge to set up and maintain a successful composting program.
- PC12.** show how to learn about bio-gasification, a technology that converts organic waste into biogas for energy production.

Waste to Energy

To be competent, the user/individual on the job must be able to:

- PC13.** explain basic concepts and types of solid waste incineration.
- PC14.** discuss recovering energy from solid waste.

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- PC15.** identify different types of incineration: mass burn, refuse-derived fuel, plasma gasification, etc.
- PC16.** perform the basic process of waste burning, ash generation, flue gas treatment, and heat recovery.
- PC17.** show how to weigh the pros and cons: reduced waste volume, energy generation, potential air pollution, and ash management challenges.

Sanitary Landfilling

To be competent, the user/individual on the job must be able to:

- PC18.** discuss concepts of solid waste landfills.
- PC19.** explain types of landfills and landfill design.
- PC20.** discuss gas movement, control and use in landfills.
- PC21.** explain leachate control and treatment in landfills.
- PC22.** perform operation and closure of landfills
- PC23.** demonstrate the basics of landfill technology, including containment, decomposition, and environmental impact.
- PC24.** show about different landfill types (e.g., sanitary, secure) and design considerations (e.g., liners, leachate collection) for safe and efficient waste disposal.
- PC25.** demonstrate gas movement and control equipped to manage landfill gas (methane) emissions through techniques like flaring, extraction, and energy production, mitigating climate impact.
- PC26.** show how to understand how to collect and treat leachate (contaminated liquid) to prevent groundwater pollution and protect ecosystems.
- PC27.** perform proper landfill operation practices and closure techniques for minimizing environmental hazards and maximizing post-closure land use potential.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** composting involves controlled decomposition of organic waste (food scraps, yard trimmings) by microorganisms under specific conditions (moisture, oxygen, temperature). It creates nutrient-rich compost, improving soil health and fertility.
- KU2.** learn about the browns (carbon-rich) and greens (nitrogen rich) materials needed for balanced composting.
- KU3.** understand different composting methods (bin, pile, vermicomposting) and choose the best for your needs. Monitor moisture, temperature, and aeration for optimal decomposition.
- KU4.** recycling involves collecting and processing specific materials (paper, plastic, glass, metal) into new products. It conserves resources, reduces energy use, and minimizes landfill waste.
- KU5.** learn about local recycling guidelines and what materials are accepted in your area. Rinse and sort recyclables properly to avoid contamination.
- KU6.** understand different recycling symbols and their meanings.
- KU7.** WtE facilities burn non-recyclable waste to generate electricity or heat. It reduces landfill waste and produces energy, but can have air quality concerns.

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- KU8.** learn about the types of waste accepted at WtE facilities in your area. Be aware of potential air emissions and their regulations, consider the environmental impact of WtE compared to other waste management options.
- KU9.** landfills are engineered disposal sites for waste that cannot be composted, recycled, or converted to energy. They aim to minimize environmental impact through liners, leachate collection, and gas capture.
- KU10.** landfills should be a last resort for waste disposal. Be aware of potential environmental concerns like methane emissions and groundwater contamination. Support policies that promote waste reduction and diversion from landfills.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** understanding the priority order of waste management (reduce, reuse, recycle, compost, waste-to-energy, landfill) is crucial.
- GS2.** analyze waste streams, identify opportunities for improvement, and adapt strategies to specific situations.
- GS3.** effectively communicate waste management plans and procedures to stakeholders and collaborate with different teams (operations, logistics, marketing).
- GS4.** analyze data on waste composition, diversion rates, and costs to evaluate progress and make informed decisions.
- GS5.** understand and comply with relevant environmental regulations and policies impacting waste management.
- GS6.** educate communities about waste management practices and encourage participation in recycling and composting programs.
- GS7.** know the science behind microbial decomposition and factors affecting compost quality.
- GS8.** identify suitable organic materials, manage moisture and aeration levels, and prevent contamination.
- GS9.** implement various composting methods (windrow, vermicomposting, in-vessel) depending on resources and scale.
- GS10.** analyze compost maturity and nutrient content to ensure compliance with standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Composting and recycling</i>	-	46	-	-
PC1. show types of solid waste recyclable materials.	-	3	-	-
PC2. prepare design and implementation of a recycling programme.	-	3	-	-
PC3. demonstrate basic concepts of solid waste composting.	-	3	-	-
PC4. show composting technologies and approaches.	-	3	-	-
PC5. show how to develop a solid waste composting programme.	-	3	-	-
PC6. perform Bio-gasification of solid waste.	-	2	-	-
PC7. show how to master the different types of recyclable materials like paper, metal, glass, plastic, and organics.	-	4	-	-
PC8. show how to learn how to implement an effective recycling program in your home, community, or organization.	-	5	-	-
PC9. ensure understanding the basics principles of composting, the breakdown of organic waste into nutrient-rich soil amendment.	-	5	-	-
PC10. show how to discover various composting methods like bin composting, vermicomposting, and windrow composting.	-	5	-	-
PC11. show how to equip yourself with the knowledge to set up and maintain a successful composting program.	-	5	-	-
PC12. show how to learn about bio-gasification, a technology that converts organic waste into biogas for energy production.	-	5	-	-
<i>Waste to Energy</i>	10	9	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. explain basic concepts and types of solid waste incineration.	5	-	-	-
PC14. discuss recovering energy from solid waste.	5	-	-	-
PC15. identify different types of incineration: mass burn, refuse-derived fuel, plasma gasification, etc.	-	3	-	-
PC16. perform the basic process of waste burning, ash generation, flue gas treatment, and heat recovery.	-	3	-	-
PC17. show how to weigh the pros and cons: reduced waste volume, energy generation, potential air pollution, and ash management challenges.	-	3	-	-
<i>Sanitary Landfilling</i>	20	15	-	-
PC18. discuss concepts of solid waste landfills.	3	-	-	-
PC19. explain types of landfills and landfill design.	3	-	-	-
PC20. discuss gas movement, control and use in landfills.	4	-	-	-
PC21. explain leachate control and treatment in landfills.	5	-	-	-
PC22. perform operation and closure of landfills	5	-	-	-
PC23. demonstrate the basics of landfill technology, including containment, decomposition, and environmental impact.	-	3	-	-
PC24. show about different landfill types (e.g., sanitary, secure) and design considerations (e.g., liners, leachate collection) for safe and efficient waste disposal.	-	3	-	-
PC25. demonstrate gas movement and control equipped to manage landfill gas (methane) emissions through techniques like flaring, extraction, and energy production, mitigating climate impact.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. show how to understand how to collect and treat leachate (contaminated liquid) to prevent groundwater pollution and protect ecosystems.	-	3	-	-
PC27. perform proper landfill operation practices and closure techniques for minimizing environmental hazards and maximizing post-closure land use potential.	-	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4074
NOS Name	Perform Composting and recycling, waste to energy and sanitary landfilling
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4075: Introduce climate change in context to waste.

Description

This unit tells about Introduce climate change in context to waste

Scope

The scope covers the following :

- Understand climate impact of waste and benefits of various waste management activities and identify the some strategy framework on waste management and climate change mitigation.

Elements and Performance Criteria

Waste and climate change

To be competent, the user/individual on the job must be able to:

- PC1.** discuss climate impact of waste.
- PC2.** explain strategic frameworks for climate change management.
- PC3.** explore the role of landfills as sources of methane, a potent greenhouse gas, through practical demonstrations or case studies.
- PC4.** develop practical strategies for promoting waste prevention, reuse, and recycling in communities, businesses, and industries.
- PC5.** examine the impact of climate change on vulnerable regions, such as coastal areas and low-lying islands, and explore how changes in weather patterns and sea levels can affect waste management infrastructure and practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** consume less, choose durable products, and repair instead of replacing.
- KU2.** diverting waste from landfills reduces greenhouse gas emissions and conserves resources.
- KU3.** reducing reliance on fossil fuels for powering waste management processes.
- KU4.** exploring alternatives to landfills and incineration, like waste-to-energy technologies with lower emissions.
- KU5.** implement strategies like waste reduction, reuse, recycling, composting, and efficient waste-to-energy technologies to minimize greenhouse gas emissions and resource depletion.
- KU6.** encourage responsible purchasing habits, product lifecycles, and end-of-life management to reduce waste generation and its associated climate impacts.
- KU7.** support policies that incentivize waste reduction, promote circular economies, and invest in sustainable waste management infrastructure.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** a strong understanding of the science and impacts of climate change is essential for anyone working on waste reduction and management. This includes knowledge of greenhouse gas emissions, the role of landfills and incineration, and the circular economy.
- GS2.** the waste sector is complex and interconnected, so being able to think critically and creatively to solve problems is essential. This includes identifying root causes of waste generation, evaluating different waste management options, and developing innovative solutions.
- GS3.** effectively communicating the importance of waste reduction and management to diverse audiences is key to raising awareness and driving change.
- GS4.** data plays a vital role in understanding waste generation patterns, measuring the impact of waste management interventions, and making informed decisions, being able to analyze and interpret data is a valuable skill.
- GS5.** depending on your specific role, you may need expertise in different areas of waste management, such as composting, recycling, anaerobic digestion, or waste-to-energy technologies.
- GS6.** many waste reduction and management projects are complex and require strong project management skills to ensure they are delivered on time, within budget, and to the desired outcomes.
- GS7.** understanding the policy and regulatory landscape around waste management is important for advocating for change and ensuring compliance with relevant regulations.
- GS8.** life cycle assessment (LCA) is a tool used to assess the environmental impacts of products and services throughout their entire life cycle.
- GS9.** Understanding LCA can help you make more informed decisions about waste management options.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Waste and climate change</i>	20	30	-	-
PC1. discuss climate impact of waste.	10	-	-	-
PC2. explain strategic frameworks for climate change management.	10	-	-	-
PC3. explore the role of landfills as sources of methane, a potent greenhouse gas, through practical demonstrations or case studies.	-	10	-	-
PC4. develop practical strategies for promoting waste prevention, reuse, and recycling in communities, businesses, and industries.	-	10	-	-
PC5. examine the impact of climate change on vulnerable regions, such as coastal areas and low-lying islands, and explore how changes in weather patterns and sea levels can affect waste management infrastructure and practices.	-	10	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4075
NOS Name	Introduce climate change in context to waste.
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4076: Plan and develop strategy for waste management focusing circular economy and plastic waste management

Description

This unit tells about how to plan and develop strategy for waste management focusing circular economy and plastic waste management

Scope

The scope covers the following :

- Learn about Greening the waste sector and Concepts related to 4R
- Perform characterisation of plastic waste and plastic waste processing technologies and plastic waste management and handling rules.

Elements and Performance Criteria

Strategy development and planning for Waste Management

To be competent, the user/individual on the job must be able to:

- PC1.** explain solid waste management planning process.
- PC2.** discuss financial, institutional, political, social and legal aspects.
- PC3.** discuss about integrated solid waste management system.
- PC4.** explain seven-step approach to developing a plan.
- PC5.** show how to understand the 7-step process, from data collection to implementation.
- PC6.** show how to consider financial, institutional, political, social, and legal aspects.
- PC7.** show how to choose the right combination of reduction, reuse, recycling, treatment, and disposal methods.
- PC8.** show how to analyze data, identify challenges, and choose sustainable solutions.
- PC9.** show how to work with stakeholders to develop and implement the plan.

Circular Economy (formerly known as "Towards Zero Waste Society")

To be competent, the user/individual on the job must be able to:

- PC10.** demonstrate concepts related to 4R.
- PC11.** demonstrate role of partnerships towards resource efficiency.
- PC12.** show how to greening the waste sector.

Plastic waste in solid waste

To be competent, the user/individual on the job must be able to:

- PC13.** introduce about polymers.
- PC14.** explain need of plastics in society.
- PC15.** explain plastic waste.
- PC16.** discuss about characterisation of plastic waste.
- PC17.** explain problems related to plastic waste.
- PC18.** explain guidelines for disposal of plastic waste.

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- PC19.** discuss about plastic waste management.
- PC20.** explain uses of recycled plastic waste.
- PC21.** discuss plastic tar road laying.
- PC22.** show how to reduce or avoid single-use plastics and informed choices on plastic use in daily life.
- PC23.** show how to properly dispose of plastics for recycling and responsible waste management.
- PC24.** show how to understand the potential of recycled plastic in applications like road construction.
- PC25.** show how to promote awareness and responsible plastic use in your community.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** implement efficient collection systems for different waste streams (compostable, recyclables, etc.).
- KU2.** understand the composition and quantity of waste generated.
- KU3.** align with circular economy principles (e.g., reduce landfill waste by 50 percentage).
- KU4.** develop action plans responsible parties, and timelines for each goal.
- KU5.** waste management systems can be designed to facilitate reuse, recycling, and resource recovery.
- KU6.** analyze your local waste composition data to understand the types and quantities of waste generated.
- KU7.** identify relevant environmental laws, waste management regulations, and extended producer responsibility (EPR) schemes.
- KU8.** involve communities, businesses, and other stakeholders in the planning process.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze existing waste management systems, identify challenges, and develop solutions that are efficient, sustainable, and inclusive.
- GS2.** work effectively with diverse stakeholders, including government agencies, businesses, NGOs, and communities, to build consensus and support.
- GS3.** develop and implement strategic plans, set goals, manage resources, and track progress towards objectives.
- GS4.** analyze data on waste generation, composition, and management to inform decision-making.
- GS5.** understand different waste collection, treatment, and disposal technologies and their environmental and economic impacts
- GS6.** apply circular economy principles like reduce, reuse, recycle, and recover to design waste management systems that minimize waste generation and maximize resource recovery.
- GS7.** understand relevant laws and regulations governing waste management and how they can be used to incentivize sustainable practices.



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- GS8.** evaluate the cost-effectiveness of different waste management options and secure funding for implementing strategies.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Strategy development and planning for Waste Management</i>	16	20	-	-
PC1. explain solid waste management planning process.	4	-	-	-
PC2. discuss financial, institutional, political, social and legal aspects.	4	-	-	-
PC3. discuss about integrated solid waste management system.	4	-	-	-
PC4. explain seven-step approach to developing a plan.	4	-	-	-
PC5. show how to understand the 7-step process, from data collection to implementation.	-	4	-	-
PC6. show how to consider financial, institutional, political, social, and legal aspects.	-	4	-	-
PC7. show how to choose the right combination of reduction, reuse, recycling, treatment, and disposal methods.	-	4	-	-
PC8. show how to analyze data, identify challenges, and choose sustainable solutions.	-	4	-	-
PC9. show how to work with stakeholders to develop and implement the plan.	-	4	-	-
<i>Circular Economy (formerly known as "Towards Zero Waste Society")</i>	-	14	-	-
PC10. demonstrate concepts related to 4R.	-	4	-	-
PC11. demonstrate role of partnerships towards resource efficiency.	-	5	-	-
PC12. show how to greening the waste sector.	-	5	-	-
<i>Plastic waste in solid waste</i>	34	16	-	-
PC13. introduce about polymers.	3	-	-	-
PC14. explain need of plastics in society.	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. explain plastic waste.	4	-	-	-
PC16. discuss about characterisation of plastic waste.	4	-	-	-
PC17. explain problems related to plastic waste.	4	-	-	-
PC18. explain guidelines for disposal of plastic waste.	4	-	-	-
PC19. discuss about plastic waste management.	4	-	-	-
PC20. explain uses of recycled plastic waste.	4	-	-	-
PC21. discuss plastic tar road laying.	4	-	-	-
PC22. show how to reduce or avoid single-use plastics and informed choices on plastic use in daily life.	-	4	-	-
PC23. show how to properly dispose of plastics for recycling and responsible waste management.	-	4	-	-
PC24. show how to understand the potential of recycled plastic in applications like road construction.	-	4	-	-
PC25. show how to promote awareness and responsible plastic use in your community.	-	4	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4076
NOS Name	Plan and develop strategy for waste management focusing circular economy and plastic waste management
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4077: Perform E-Waste Management

Description

This unit tells about Perform E-Waste Management

Scope

The scope covers the following :

- Discuss about the methodology of e-waste inventory and explain the importance of Conceptual approaches and methodology.
- Explain component of E-waste management and explain laws regulations and policies of E-waste.

Elements and Performance Criteria

E-waste - 1

To be competent, the user/individual on the job must be able to:

- PC1.** introduce E-waste definition
- PC2.** discuss guidelines for assessment of E-waste market.
- PC3.** discuss guidelines for selection of methodology for e-waste inventory.
- PC4.** discuss guidelines for e-waste inventory assessment.
- PC5.** discuss conceptual approaches and methodology.
- PC6.** explain any Case study
- PC7.** define e-waste accurately.
- PC8.** assess the e-waste market in your region using established guidelines.
- PC9.** explain how to select the appropriate methodology for conducting an e-waste inventory based on specific needs and context.
- PC10.** discuss how to implement various e-waste inventory assessment techniques.
- PC11.** explain how to apply conceptual approaches and methodologies to real-world scenarios through a case study.

E-waste - 2

To be competent, the user/individual on the job must be able to:

- PC12.** describe introduction and perspectives of E-waste management.
- PC13.** discuss current practices of E-waste management.
- PC14.** discuss laws, regulations and policies on E-waste.
- PC15.**
 - discuss E-waste management – stages and technologies
 - Financing mechanism.
- PC16.** explain any one case study.
- PC17.** identify e-waste items, their hazardous components, and potential environmental/health impacts.
- PC18.** perform local e-waste practices on collection/recycling options in your area.
- PC19.** show how to understand key regulations and producer responsibility policies influencing e-waste management.

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- PC20.** explore different e-waste processing methods (disassembly, recycling, etc.) and their benefits/limitations.
- PC21.** identify potential funding sources for effective e-waste management initiatives.
- PC22.** show how to apply learnings to real-world examples, evaluating successes and challenges in implementing best practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding the different types of materials present in various electronics, their potential hazards, and their recyclability/reusability.
- KU2.** understanding of the harmful consequences of improper e-waste disposal, including air and water pollution, health risks to workers and communities, and resource depletion.
- KU3.** knowledge of local, national, and international regulations governing e-waste management, including proper handling, storage, transportation, and disposal practices.
- KU4.** understanding different collection methods (individual drop-off, organized campaigns), logistics involved in safe transportation, and responsible partnerships with recyclers.
- KU5.** knowledge of various recycling technologies for different materials, safe dismantling techniques, and responsible downstream processing chains.
- KU6.** awareness of the importance of data wiping and destruction before processing e-waste to protect sensitive information.
- KU7.** understanding the role of consumer education in promoting responsible e-waste disposal habits, including take-back programs and product-life cycle awareness.
- KU8.** knowledge of industry initiatives and partnerships for responsible e-waste management, extended producer responsibility schemes, and best practices sharing.
- KU9.** understanding the importance of building awareness and capacity within communities to participate in safe and responsible e-waste management practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** E-waste often contains hazardous materials, so it's important to be able to carefully sort and handle it according to regulations.
- GS2.** you'll need to be able to communicate effectively with colleagues, clients, and regulators.
- GS3.** you'll need to be familiar with the laws and regulations that govern e-waste management in your area.
- GS4.** you'll need to know how e-waste is processed and recycled.
- GS5.** understanding the various local, national, and international regulations that govern the collection, transportation, treatment, and disposal of e-waste.
- GS6.** ability to recognize different types of electronic equipment and their components, and to understand the potential hazards associated with each type.
- GS7.** understanding the different methods used to recycle, reuse, or dispose of e-waste, such as dismantling, shredding, and smelting.



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- GS8.** knowledge how to safely store, transport, and process e-waste to avoid harm to human health and the environment.
- GS9.** ability to clearly and concisely communicate technical information to both technical and non-technical audiences.
- GS10.** able to collect, analyze, and interpret data to track progress, identify trends, and make informed decisions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>E-waste - 1</i>	45	-	-	-
PC1. introduce E-waste definition	3	-	-	-
PC2. discuss guidelines for assessment of E-waste market.	4	-	-	-
PC3. discuss guidelines for selection of methodology for e-waste inventory.	4	-	-	-
PC4. discuss guidelines for e-waste inventory assessment.	3	-	-	-
PC5. discuss conceptual approaches and methodology.	3	-	-	-
PC6. explain any Case study	3	-	-	-
PC7. define e-waste accurately.	5	-	-	-
PC8. assess the e-waste market in your region using established guidelines.	5	-	-	-
PC9. explain how to select the appropriate methodology for conducting an e-waste inventory based on specific needs and context.	5	-	-	-
PC10. discuss how to implement various e-waste inventory assessment techniques.	5	-	-	-
PC11. explain how to apply conceptual approaches and methodologies to real-world scenarios through a case study.	5	-	-	-
<i>E-waste - 2</i>	25	30	-	-
PC12. describe introduction and perspectives of E-waste management.	5	-	-	-
PC13. discuss current practices of E-waste management.	5	-	-	-
PC14. discuss laws, regulations and policies on E-waste.	5	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. • discuss E-waste management – stages and technologies • Financing mechanism.	5	-	-	-
PC16. explain any one case study.	5	-	-	-
PC17. identify e-waste items, their hazardous components, and potential environmental/health impacts.	-	5	-	-
PC18. perform local e-waste practices on collection/recycling options in your area.	-	5	-	-
PC19. show how to understand key regulations and producer responsibility policies influencing e-waste management.	-	5	-	-
PC20. explore different e-waste processing methods (disassembly, recycling, etc.) and their benefits/limitations.	-	5	-	-
PC21. identify potential funding sources for effective e-waste management initiatives.	-	5	-	-
PC22. show how to apply learnings to real-world examples, evaluating successes and challenges in implementing best practices.	-	5	-	-
NOS Total	70	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4077
NOS Name	Perform E-Waste Management
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N4078: Perform Construction and Demolition activities within waste management frameworks, alongside addressing Bio-Medical Waste Management concerns and explores the fundamental principles of Geographic Information Systems applied to waste management practices.

Description

This unit tells about perform Construction and Demolition in Waste Management. Introduce about Bio-Medical Waste Management and fundamentals of Geographical Information system in waste Management. The course will facilitate management of various categories of waste (e.g., solid waste, plastic, biomedical waste, e-waste, C&D waste etc.) including source segregation, collection, disposal, processing and recycling.

Scope

The scope covers the following :

- Learn about components of construction and demolition waste and explain the importance of C & D waste management.
- Understand the Characteristics, & source of Biomedical waste and identify career opportunities in Bio-Medical waste management planning.
- Understand about the fundamental of GIS and explain the importance of GIS software

Elements and Performance Criteria

Construction and Demolition Waste Management

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about construction and demolition waste.
- PC2.** explain construction and demolition waste rules 2016.
- PC3.** discuss inventorisation of Construction and demolition waste.
- PC4.** explain Collection, transport and disposal
- PC5.** discuss about processing and utilization.
- PC6.** show how to reduce your C&D waste footprint and contribute to a more sustainable construction industry.
- PC7.** show how to comply with relevant regulations and avoid potential penalties.
- PC8.** show how to optimize costs associated with waste management.
- PC9.** show how to promote resource recovery and circular economy principles in the construction sector.

Bio-Medical Waste Management

To be competent, the user/individual on the job must be able to:

- PC10.** explain definition, characteristics and sources of Bio-Medical waste
- PC11.** discuss about risks associated.
- PC12.** discuss legislative, regulatory and policy aspect.
- PC13.** discuss Bio-Medical waste management planning.

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- PC14.** discuss waste minimization, reuse and recycling.
- PC15.** discuss about classification, segregation, storage and transport.
- PC16.** describe treatment and disposal methods.
- PC17.** describe collection and disposal of waste water.
- PC18.** discuss health and safety practices.
- PC19.** discuss about hospital hygiene and infection control.
- PC20.** discuss bio-Medical waste management in emergencies.
- PC21.** discuss issues and challenges.
- PC22.** minimize risks: Safely handle, classify, segregate, store, transport, and dispose of bio-medical waste to prevent infections, environmental harm, and occupational hazards.
- PC23.** show how to implement waste management plans within legal and policy frameworks, ensuring responsible and sustainable practices.
- PC24.** show how to employ techniques like segregation and waste minimization to decrease the amount of bio-medical waste generated.
- PC25.** show how to choose appropriate treatment and disposal methods based on waste type and risk, considering environmental impact and cost-effectiveness.
- PC26.** show how to implement proper hygiene and infection control measures to safeguard healthcare workers and the community from exposure to bio-medical waste.
- PC27.** show how to adapt waste management practices during crises and disasters to maintain safety and sanitation.
- PC28.** show how to address potential issues like resource constraints, lack of awareness, and inadequate infrastructure.

GIS

To be competent, the user/individual on the job must be able to:

- PC29.** introduce GIS
- PC30.** explain GIS Software's.
- PC31.** discuss fundamental of GIS.
- PC32.** describe layers.
- PC33.** discuss spatial Data and GIS.
- PC34.** discuss survey methods.
- PC35.** explain fundamental of maps.
- PC36.** discuss resource mapping exercises.
- PC37.** explain use of GIS and GPS tools and applications for data collection and mapping.
- PC38.** operate popular GIS software like ArcGIS or QGIS to create and manage geographic information.
- PC39.** show how to understand the core principles of GIS layers, spatial data, projections, and coordinate systems.
- PC40.** show how to collect accurate spatial data using various survey methods and GPS technology.
- PC41.** show how to interpret and analyze maps effectively, extracting valuable insights from geographic patterns.
- PC42.** show how to conduct resource mapping exercises to identify, measure, and monitor natural resources like water, forests, or minerals.

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- PC43.** apply GIS and GPS tools in real-world scenarios for data collection, mapping, and resource management.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding of concrete, bricks, wood, metal, glass, asphalt, gypsum, and potentially hazardous materials like asbestos and lead.
- KU2.** knowledge local regulations and permits governing C&D waste handling, storage, and disposal.
- KU3.** techniques for separating different materials at the source to maximize recycling and minimize landfilling.
- KU4.** crushing concrete for base material, reusing bricks, converting wood into mulch, etc.
- KU5.** identifying and addressing safety hazards associated with demolition and waste handling.
- KU6.** landfilling C&D waste contributes to resource depletion, air and water pollution, and ecosystem damage.
- KU7.** effective C&D waste management saves resources, reduces costs, and promotes sustainability.
- KU8.** knowledge of types of bio-medical waste like infectious waste, sharps, pathological waste, animal waste, chemical waste, etc.
- KU9.** improper bio-medical waste management poses significant health risks to healthcare workers, waste handlers, and the community.
- KU10.** untreated bio-medical waste can contaminate water resources and soil, posing ecological risks.
- KU11.** using GIS to map waste generation, collection routes, disposal sites, and environmental sensitivities.
- KU12.** generating maps and infographics to raise public awareness about waste management issues.
- KU13.** creating clear and effective maps and visualizations to communicate waste management challenges and solutions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to analyze waste data to identify trends and improve efficiency.
- GS2.** planning, organizing, and executing C&D waste management projects.
- GS3.** ability to differentiate reusable, recyclable, and non-recyclable materials.
- GS4.** knowledge of local, state, and federal laws and regulations governing construction and demolition waste disposal.
- GS5.** understanding of the types of waste that are generated and the challenges involved in managing it.
- GS6.** knowledge of waste management best practices like source reduction, recycling, and composting.

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- GS7.** biomedical waste is subject to strict regulations, which vary depending on the jurisdiction.
- GS8.** Biomedical waste must be treated and disposed of in a way that minimizes the risk of harm to human health and the environment.
- GS9.** All personnel who handle biomedical waste must be properly trained.
- GS10.** GIS can be used to identify potential sites for landfills or other waste disposal facilities, taking into account factors such as environmental sensitivity and proximity to population centers.
- GS11.** GIS can be used to monitor compliance with waste management regulations.
- GS12.** GIS can be used to map the location and quantity of waste generation. This information can be used to identify areas where waste reduction or recycling efforts could be most effective.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Construction and Demolition Waste Management</i>	10	11	-	-
PC1. discuss about construction and demolition waste.	2	-	-	-
PC2. explain construction and demolition waste rules 2016.	2	-	-	-
PC3. discuss inventorisation of Construction and demolition waste.	2	-	-	-
PC4. explain Collection, transport and disposal	2	-	-	-
PC5. discuss about processing and utilization.	2	-	-	-
PC6. show how to reduce your C&D waste footprint and contribute to a more sustainable construction industry.	-	2	-	-
PC7. show how to comply with relevant regulations and avoid potential penalties.	-	3	-	-
PC8. show how to optimize costs associated with waste management.	-	3	-	-
PC9. show how to promote resource recovery and circular economy principles in the construction sector.	-	3	-	-
<i>Bio-Medical Waste Management</i>	22	21	-	-
PC10. explain definition, characteristics and sources of Bio-Medical waste	1	-	-	-
PC11. discuss about risks associated.	1	-	-	-
PC12. discuss legislative, regulatory and policy aspect.	2	-	-	-
PC13. discuss Bio-Medical waste management planning.	2	-	-	-
PC14. discuss waste minimization, reuse and recycling.	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. discuss about classification, segregation, storage and transport.	2	-	-	-
PC16. describe treatment and disposal methods.	2	-	-	-
PC17. describe collection and disposal of waste water.	2	-	-	-
PC18. discuss health and safety practices.	2	-	-	-
PC19. discuss about hospital hygiene and infection control.	2	-	-	-
PC20. discuss bio-Medical waste management in emergencies.	2	-	-	-
PC21. discuss issues and challenges.	2	-	-	-
PC22. minimize risks: Safely handle, classify, segregate, store, transport, and dispose of bio-medical waste to prevent infections, environmental harm, and occupational hazards.	-	3	-	-
PC23. show how to implement waste management plans within legal and policy frameworks, ensuring responsible and sustainable practices.	-	3	-	-
PC24. show how to employ techniques like segregation and waste minimization to decrease the amount of bio-medical waste generated.	-	3	-	-
PC25. show how to choose appropriate treatment and disposal methods based on waste type and risk, considering environmental impact and cost-effectiveness.	-	3	-	-
PC26. show how to implement proper hygiene and infection control measures to safeguard healthcare workers and the community from exposure to bio-medical waste.	-	3	-	-
PC27. show how to adapt waste management practices during crises and disasters to maintain safety and sanitation.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. show how to address potential issues like resource constraints, lack of awareness, and inadequate infrastructure.	-	3	-	-
<i>GIS</i>	18	18	-	-
PC29. introduce GIS	2	-	-	-
PC30. explain GIS Software's.	2	-	-	-
PC31. discuss fundamental of GIS.	2	-	-	-
PC32. describe layers.	2	-	-	-
PC33. discuss spatial Data and GIS.	2	-	-	-
PC34. discuss survey methods.	2	-	-	-
PC35. explain fundamental of maps.	2	-	-	-
PC36. discuss resource mapping exercises.	2	-	-	-
PC37. explain use of GIS and GPS tools and applications for data collection and mapping.	2	-	-	-
PC38. operate popular GIS software like ArcGIS or QGIS to create and manage geographic information.	-	3	-	-
PC39. show how to understand the core principles of GIS layers, spatial data, projections, and coordinate systems.	-	3	-	-
PC40. show how to collect accurate spatial data using various survey methods and GPS technology.	-	3	-	-
PC41. show how to interpret and analyze maps effectively, extracting valuable insights from geographic patterns.	-	3	-	-
PC42. show how to conduct resource mapping exercises to identify, measure, and monitor natural resources like water, forests, or minerals.	-	3	-	-
PC43. apply GIS and GPS tools in real-world scenarios for data collection, mapping, and resource management.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4078
NOS Name	Perform Construction and Demolition activities within waste management frameworks, alongside addressing Bio-Medical Waste Management concerns and explores the fundamental principles of Geographic Information Systems applied to waste management practices.
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Entrepreneur
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e-mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4072.Introduction to Solid Waste	25	25	0	0	50	12
SGJ/N4073.Identify characteristic, collection , transfer and transport and Separation and processing of solid waste	16	35	0	0	51	12
SGJ/N4074.Perform Composting and recycling, waste to energy and sanitary landfilling	30	70	0	0	100	13
SGJ/N4075.Introduce climate change in context to waste.	20	30	0	0	50	12
SGJ/N4076.Plan and develop strategy for waste management focusing circular economy and plastic waste management	50	50	0	0	100	13
SGJ/N4077.Perform E-Waste Management	70	30	0	0	100	13

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4078.Perform Construction and Demolition activities within waste management frameworks, alongside addressing Bio-Medical Waste Management concerns and explores the fundamental principles of Geographic Information Systems applied to waste management practices.	50	50	0	0	100	13
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	12
Total	281	320	0	0	601	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.